

SIDDHANT PODDAR

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PROFESSIONAL SUMMARY

Computer Science (Cybersecurity) undergraduate with strong foundations in application security, network defense, cryptography, and secure backend engineering. Experienced in developing zero-trust API security gateways, automated Linux administration tools, and database-driven systems using Python, Java, SQL, and FastAPI. Proficient in threat modeling, vulnerability identification (OWASP Top 10), and packet inspection, with a demonstrated ability to translate complex security requirements into reliable, high-performance software solutions.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, Java, C, JavaScript, SQL, Bash / Shell Scripting
- **Cybersecurity & Networking:** Application Security (OWASP Top 10), Network Security, Cryptography (AES, RSA, HMAC-SHA256), Zero-Trust Architecture, Role-Based Access Control (RBAC), TCP/IP Protocols, Packet Analysis
- **Security Tools & Platforms:** Wireshark, Burp Suite, Nmap, Linux/Unix Security, Postman, Git, GitHub, VS Code
- **Backend & Architecture:** FastAPI, RESTful APIs, Asynchronous I/O (asyncio), Pydantic, Object-Oriented Design, System Architecture
- **Cloud & Databases:** AWS Fundamentals (IAM, VPC, S3), MySQL, PostgreSQL, In-Memory Data Structures, Secure API Gateways

FEATURED PROJECTS

AegisPay-AI — Zero-Trust Security Mesh & Payment Firewall for AI Agents

FLAGSHIP

GitHub: github.com/sidpoddar07/AegisPay-AI | Tech Stack: Python, FastAPI, Pydantic, React, Vite, Cryptography, Pytest

- Engineered a high-throughput, in-flight Zero-Trust security gateway that intercepts autonomous AI agent payment requests before touching payment gateway APIs (e.g., Razorpay).
- Designed an **In-Flight Cascaded Architecture** achieving a mean latency of **0.036 ms** (< 1ms SLA) by combining pre-compiled regex automata, Shannon entropy obfuscation detection, and bidirectional intent divergence algorithms.
- Implemented a **Cryptographic Proof-of-Intent (PoI) Token** using time-bound (60s TTL) HMAC-SHA256 to eliminate in-flight Man-In-The-Middle (MITM) parameter tampering and amount inflation attacks.
- Developed a 3-tier composite risk matrix with an adaptive **REVIEW band** (Risk 30–74) to trigger Step-Up 2FA challenges, reducing false-positive checkout drops.
- Built an automated regulatory compliance engine that synthesizes **RBI-AML/PMLA-compliant Suspicious Activity Reports (SAR)** sealed with SHA-256 tamper-evident integrity hashes.
- Achieved **100% test coverage (32/32 passing tests)** and created a real-time Cyber SOC Dashboard with live penetration attack simulation.

Smart Attendance Management System

Tech Stack: Python, Web Technologies, MySQL, RBAC, REST APIs

- Developed a secure web-based attendance automation platform reducing manual operational overhead by 80%.
- Implemented robust session-based authentication, Role-Based Access Control (RBAC), and parameterized SQL queries to safeguard student data against injection attacks.
- Architected a centralized reporting dashboard with optimized indexing for high-speed attendance record retrieval.

Linux Automation & Infrastructure Toolkit

Tech Stack: Bash, Shell Scripting, Linux System Administration, Cron

- Engineered modular Bash automation utilities for automated log rotation, resource threshold alerting, and background task scheduling.
- Implemented defensive scripting patterns with automated error-handling traps, permission auditing, and secure file integrity checks.

EDUCATION

B.Tech in Computer Science (Cyber Security) | Dayananda Sagar University, Bangalore

2023 – 2027

Higher Secondary Certificate (12th) | Kendriya Vidyalaya

2020 – 2022 | Score: 74.20%

CERTIFICATIONS & TRAININGS

- IBM Ethical Hacking Principles | • Fundamentals of Information Security | • Fundamentals of Cryptography
- CompTIA Security+ (In Progress) | • AWS Cloud Practitioner (In Progress)